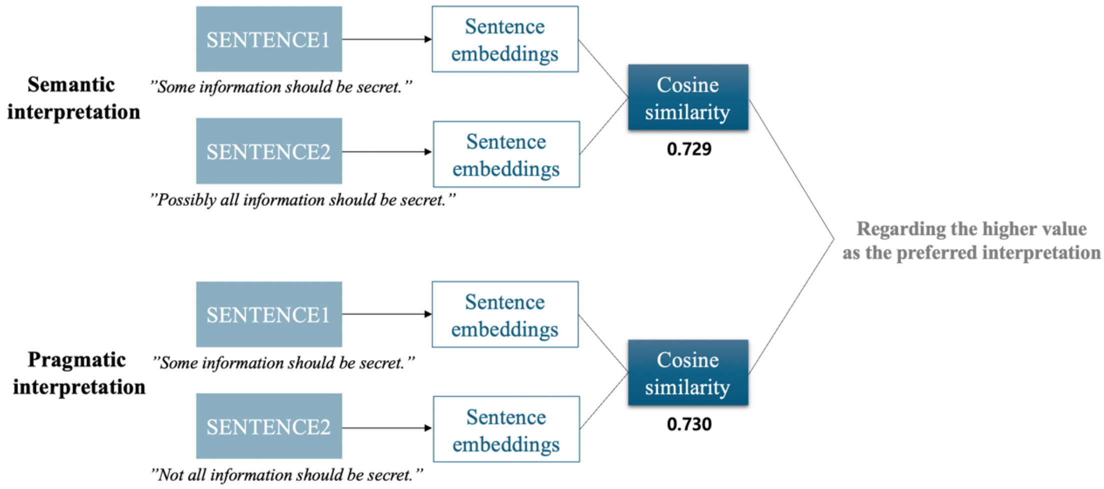

Figure 1. Overview of embedding some-sentences and its semantic and pragmatic counterparts, measuring cosine similarities between SENTENCE1 and SENTENCE2, and selecting the models' preferred interpretation between semantic and pragmatic interpretations in experiment 1

1. Do LLMs perform pragmatic interpretation rather than semantic interpretation for scalar implicature without context?
2. Do LLMs exhibit sensitivity to a contextual cue, such as QUD, in discourse during the processing of scalar implicature?

To address these questions, we will conduct two experiments in the following sections.

3 Data collection

To investigate the processing of scalar implicature by LLMs, we extracted sentences with '*some* + NP' structures from British National Corpus (BNC) using NLTK (Bird, 2006). Among those, we collected sentences where '*some* + NP' was positioned as the subject due to the fact that the implicature generation is stronger when '*some* + NP' is positioned at the sentence-initial position compared to the sentence-final position (Breheny 2006). In addition, we excluded sentences with multiple clauses to avoid the possibility of cancellation. Finally, a total of 198 sentences were extracted and one example of the final data is presented as in (9).

(9) Some information should be secret.
(BNC W:newsp:other:social, K5C-156)

SENTENCE1	SENTENCE2	Interpretation
Some information should be secret.	**Possibly all** information should be secret.	Semantic
Some information should be secret.	**Not all** information should be secret.	Pragmatic

Table1. Materials for experiment 1

We refer to the sentences extracted through this process as *some*-sentences. Both data and results of the experiments are publicly available.[1]

4 Experiment 1

Previous experiments on human language processing have successfully captured pragmatic inference of scalar implicature even without context (Noveck and Posada, 2003; De Neys and Schaeken, 2007; Huang and Snedeker, 2009; Hunt et al., 2013; Tomlinson et al., 2013). Likewise, experiment 1 aimed to investigate how LLMs interpret *some*-sentences without context, distinguishing between semantic entailment or pragmatic implicature.

4.1 Method

The experimental materials consisted of *some*-sentences and sentences with its semantic and pragmatic interpretations as shown in Table 1.

SENTENCE1 was composed of the *some*-sentences, while SENTENCE2 included sentences with either semantic or pragmatic interpretations. To ensure uniform token count between the two

[1] https://github.com/joyennn/scalar-implicature

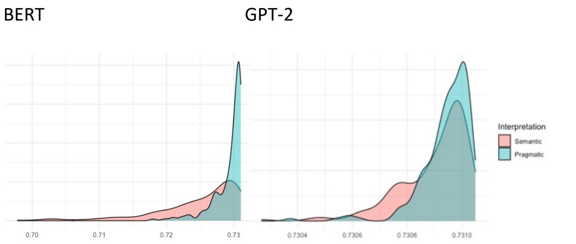

Figure 2. Density of cosine similarities between *some*-sentences and its semantic or pragmatic interpretations

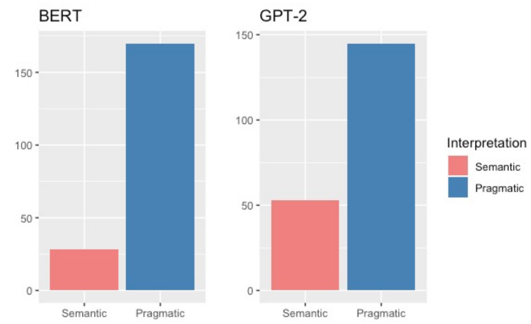

Figure 3. Instances with higher similarities for the same *some*-sentences across interpretations

sentences in SENTENCE2, the sentence with the semantic interpretation used only *possibly all* instead of *at least one and possibly all*. Each pair of SENTENCE1 and SENTENCE2 was labeled as either 'Semantic' or 'Pragmatic' depending on its interpretation.

LLMs used for the experiment were BERT (Devlin et al., 2019) and GPT-2 (Radford et al., 2019), both of which were transformers-based pre-trained language models. BERT (`BERT-base-uncased`) comprises 12 Transformer encoder layers, each of which is designed to capture bidirectional context from the input text. Unlike BERT, which uses only encoder layers, GPT-2 small (`gpt2`) utilizes 12 decoder layers to generate text in an autoregressive manner, predicting the next word in a sequence based on the previously generated words. Despite these differences, both models have a hidden size of 768 and 12 self-attention heads. In addition, the total number of parameters are similar in BERT and GPT-2 which have approximately 110 million and 117 million parameters, respectively. Although newer and more advanced models have proved higher performance, BERT and GPT-2, as foundational transformer models, are well-known in terms of their processing architectures, which allows us to better understand how these models process language.

Specifically, input sentences were tokenized using each model's tokenizer. For BERT, we obtained sentence embeddings by using the [CLS] token embeddings from the final layer. On the other hand, for GPT-2, sentence embeddings were derived by averaging the token embeddings from the final layer. We then computed the cosine similarity between pairs of corresponding sentence embeddings for SENTENCE1 and SENTENCE2. Cosine similarity is a method that measures how similar two sentences are by evaluating the angle between two sentence vectors. Although it may underestimate the similarity of words or sentences (Zhou et al., 2022), it is not just suitable for measuring the similarity of sentences but also computationally efficient and widely used in many studies. These cosine similarity scores were averaged to obtain a single similarity measure for the sentence pairs.

Since the value of cosine similarity ranges from [-1, 1], it was linearly transformed to a [0, 1] range for ease of interpretation. Then, the sigmoid function was applied to ensure to avoid values that are extremely close to 0 or 1. In this classification, a value close to 1 indicates high similarity between two sentences, while a value close to 0 indicates lower similarity. Through this metric, we measured whether the *some*-sentences were interpreted in a semantic or pragmatic manner. The overview of the experiment 1 is presented in Figure 1.

To verify statistical significance, a linear mixed-effects regression model from the lme4 package in the R statistical software was employed (Bates et al. 2014). The summaries of linear mixed-effects models are provided in the Appendix section.

4.2 Result

Figure 2 showed the density of the similarities between *some*-sentences and its semantically or pragmatically interpreted counterparts. While both interpretations exhibited similarities between 0.5 and 1, indicating high degree of sentence similarities, the pragmatic interpretations appeared relatively more prominent.

Figure 3 illustrated which interpretations, semantic or pragmatic, exhibited higher similarities for the same *some*-sentences. In BERT, 28 instances showed higher similarities to the semantic interpretations while 170 instances showed higher similarities to the pragmatic

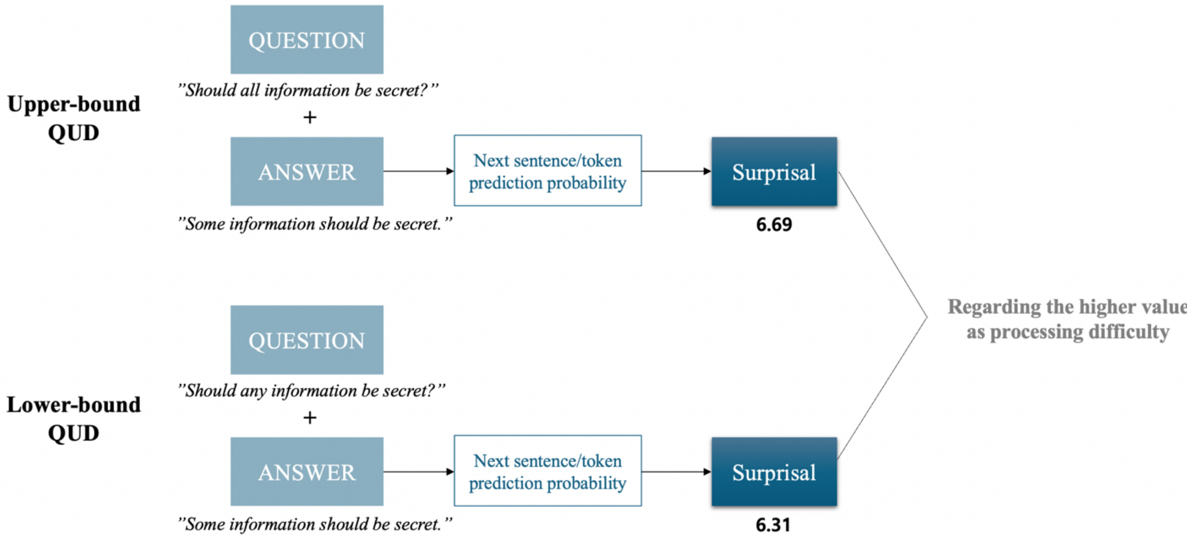

Figure 4. Overview of embedding question and answer sentences, calculating next sentence/token prediction probabilities for the answer sentences, and transforming the probabilities into surprisals in experiment 2

interpretations. In GPT-2, 53 instances exhibited higher similarities to the semantic interpretations while 145 instances exhibited higher similarities to the pragmatic interpretations. The statistical analysis revealed significant effects in the interpretations for both models ($p < 0.001$).

In summary, the interpretations of the scalar implicature *some* without context tended to be predominantly pragmatic, reflecting a consistency with human language processing.

5 Experiment 2

Building on the findings from Experiment 1, which showed that both BERT and GPT-2 models prefer pragmatic interpretations to semantic interpretations in scalar implicature without context, experiment 2 aimed to explore whether the LLMs have more processing difficulties when implicature is required (i.e., upper-bound QUD), compared to when implicature is not required (i.e., lower-bound QUD). For this comparison, the context was manipulated using QUD as a contextual cue.

5.1 Method

In the experimental materials, two types of the question sentences were generated for the *some*-sentences according to the types of QUDs, such as upper- and lower-bound QUDs. Following Yang et al. (2018), questions for the upper-bound included *all*, while those for the lower-bound included *any*.

QUESTION	ANSWER	QUD
Should **all** information be secret?	**Some** information should be secret.	Upper
Should **any** information be secret?	**Some** information should be secret.	Lower

Table2. Materials for experiment 2

As presented in Table 2, QUESTION comprised questions with either *all* or *any*, while ANSWER consisted of the *some*-sentences. Each pair of QUESTION and ANSWER was labeled as either ‘Upper’ or ‘Lower’ depending on its QUD.

In experiment 2, we also employed BERT-base and GPT-2 small models. BERT is pre-trained using Next Sentence Prediction (NSP), which involves predicting whether the second sentence immediately follows the first sentence in the given pair of sentences. This is achieved by concatenating the two sentences with [CLS] (classification start) and [SEP] (sentence separator) tokens to form the input for the BERT model. The [CLS] token embeddings from the final layer are used to compute the NSP probability, thereby quantifying the probability of ANSWER following QUESTION.

On the other hand, GPT-2 does not utilize methods like BERT’s NSP as its training data lacks explicit signals indicating relationships between sentences. Instead, GPT-2 predicts the next word based on the preceding context. To assess the relationship between two sentences in GPT-2, we combined QUESTION and ANSWER into a single